Can we use doomscrolling to help children learn?

A Chrome extension that turns short-video time into practice

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Abstract

Short videos on TikTok, YouTube Shorts and Instagram Reels make it easy to scroll for hours. Time limits only stop the video; they give nothing back and are easy to turn off. We built Cure Brainrot, a Chrome extension that closes short-video tabs after a daily limit and opens a short school quiz: 8 of 10 right answers give 20 more minutes, and the AI writes the questions for the chosen subject, grade and language. In a one-day test, 2 classmates closed 6 short-video tabs, passed 7 of 10 quizzes with 8.5 of 10 right answers on average, and rated it 4 of 5 for fun; in a hand check of 30 AI questions, 4 were unclear and none were wrong. This shows that a small quiz can turn wasted scrolling into practice, but 30 checked questions and 2 users are not enough to claim it makes children learn.



Figure 1: Our idea. When the time for short videos is over, a short school quiz comes first. (Illustration made with an image model.)

1 Introduction

The problem. Short videos eat hours without anyone noticing. My classmates and I open TikTok, YouTube Shorts or Instagram Reels "for five minutes" and lose an evening; after that it is hard to start homework or sleep. It is not only about time — the feeling after a long scroll is that the brain is tired and nothing was learned.

Why it is hard. The apps are built to keep you: an endless feed, autoplay, no natural stopping point, and a "one more video" feeling that is stronger than any plan made before opening the app. Warnings and timers alone do not work, because the easiest choice is always to ignore them or turn them off.

What we did. We made "Cure Brainrot", a Chrome extension that closes short-video tabs and lets you back in only after a short quiz. A timer counts only while a short-video tab is actually

open; when the free minutes are used, the quiz opens, and a pass gives 20 more minutes while a fail brings it back later. The quiz is a real school quiz: 10 questions, AI-made for the chosen subject, grade and screen language (Kazakh, Russian, English), and it remembers missed topics so the next quiz starts with them.
Our contributions:

- **Cure Brainrot**, a Chrome extension that counts the time on TikTok, YouTube Shorts and Instagram Reels and closes them at a daily limit;
- a school quiz (grades 5–7, English, Russian or Kazakh) that gives more time only after 8 of 10 right answers, with two sources of questions: built-in questions that a person checked, and AI questions from the class language model;
- a hand check of AI questions and a small test with 2 classmates.

2 Related work

Many TikTok users are adolescents. Montag et al. review the first studies on TikTok use and discuss how its design (an endless feed that the app chooses for you) can keep people watching [5]. In a study with 60 people, watching TikTok between tasks made it harder to remember a planned action; Twitter and YouTube did not have this effect [1]. Many apps and browser extensions already help people control their time online, for example with blocking, timers and reminders [4]. We use a different idea from learning research: a short test helps you remember more than reading the same text again (the *testing effect* [6]). So our "price" for more time is a quiz, not only a wait.

3 Cure Brainrot

Figure 2 shows how the extension works. It is a Manifest V3 Chrome extension [3] written in JavaScript. It uses the Chrome APIs `tabs`, `storage` and `alarms` [2]. Check every sentence of this section against your final version (1.8 or newer). Change what is not true any more.

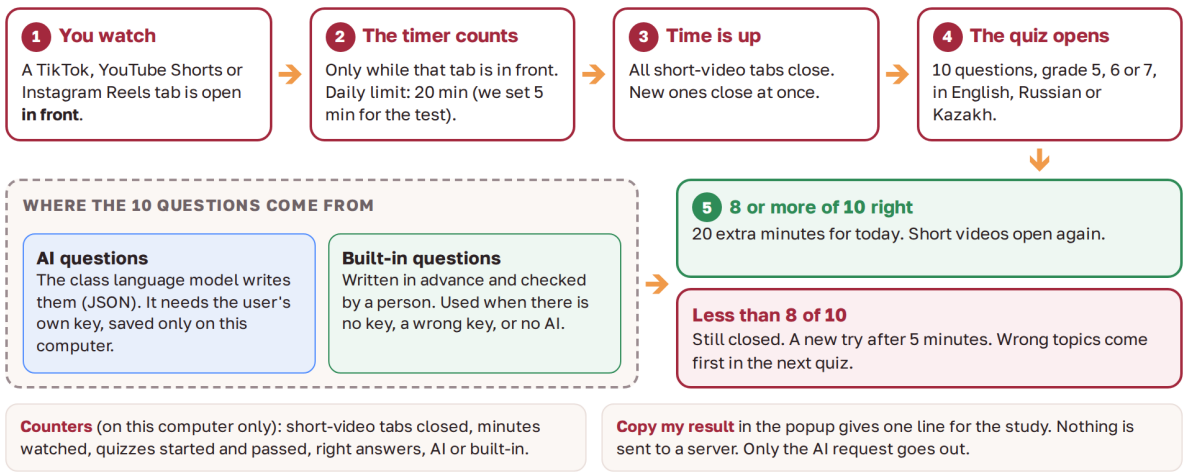


Figure 2: How Cure Brainrot works. Steps 1–4 run in the background. Step 5 depends on the quiz score.

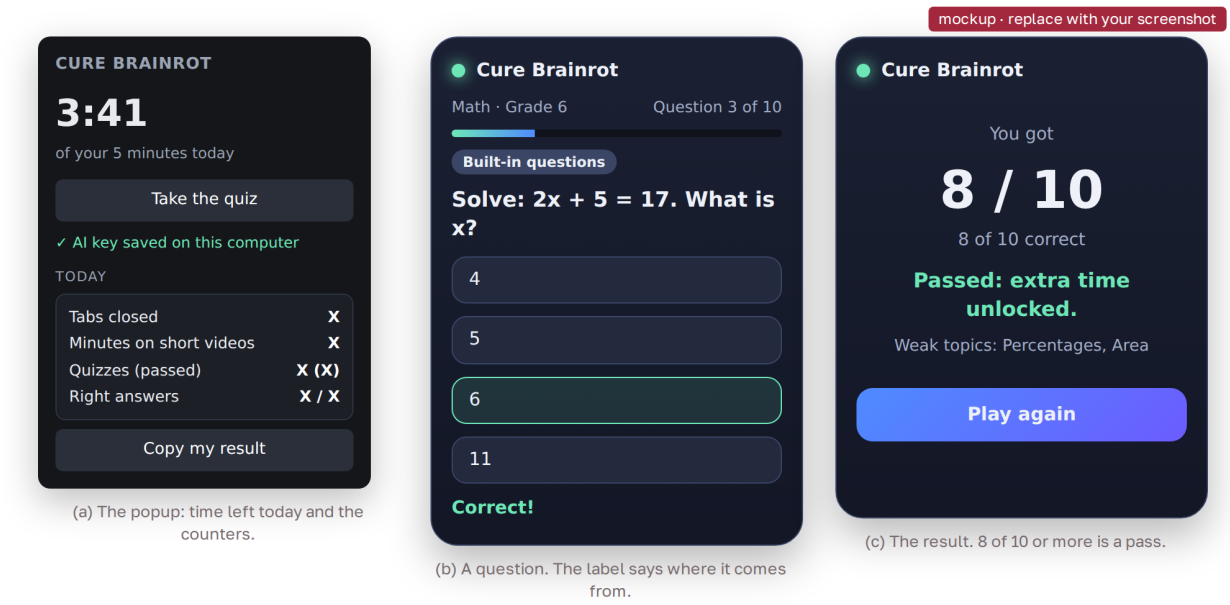
3.1 The timer and the gate

The extension watches three kinds of pages: TikTok (`tiktok.com`), YouTube Shorts (`youtube.com/shorts`) and Instagram Reels (`instagram.com/reels`). The timer counts only while one of

these pages is the active tab of the window in front, so a short-video tab in the background does not use your time. The daily limit is 20 minutes. The user can change it in the popup (1 to 600 minutes); for the test we use 5 minutes. The total starts again at 0 every day. The extension checks the time every 30 seconds and also sets an alarm for the exact moment when the limit is reached. When the time is up, the extension closes all short-video tabs, closes new ones at once, and opens the quiz page.

3.2 The quiz

At the first start the user chooses the screen language (English, Russian or Kazakh), the grade (5, 6 or 7) and one or more subjects: Math, Science and Social Studies. Users can also add their own subject; then the AI writes 6 topics for each grade. Science and Social Studies have 8 topics for each grade (for example *Photosynthesis* in grade 6 Science and *The Silk Road* in grade 6 Social Studies). A quiz has 10 multiple-choice questions with 4 options each. After each answer the quiz shows the right option (Figure 3b). The topics with wrong answers are saved as *weak topics*, and the next quiz starts with them ("Today's focus"). 8 or more right answers is a pass: the user gets 20 extra minutes for today. With fewer than 8, short videos stay closed, and a new try is possible after 5 minutes.



(a) The popup: time left today and the counters.

(b) A question. The label says where it comes from.

(c) The result. 8 of 10 or more is a pass.

Figure 3: The popup (a), a quiz question (b) and the result (c). This is a mockup in the style of our real pages: replace it with your own screenshot.

3.3 Questions

If the user picks 2 or 3 subjects, the 10 questions are shared between them. The extension keeps only questions with a question text, 4 non-empty options and an answer number from 0 to 3, and it mixes the order of the options. Each user needs a personal key. The key is saved only on the user's computer (`chrome.storage.local`), never in our code. One key can make 200 requests per day.

3.4 Privacy

The counters (Section 4) remain in the browser on the user's computer. The extension sends nothing to the server, except for the request for AI questions. That request has only the subject,

Source	Subjects (grades 5–7)	Questions
AI	Math	new every quiz
AI	Science	new every quiz
AI	Social Studies	new every quiz
AI	Chosen subject	new every quiz

Table 1: All questions can be in English, Russian or Kazakh.

the grade, the language, and the topics.

4 How we measure

4.1 Are the questions right?

When checking the question we used 10 questions from each subject for grade 6. We checked each question and marked it *good* or *unclear*. A question is wrong if the marked answer is wrong or if more than one option is right.

Questions	good	unclear
Math	10	0
Science	9	1
Social Studies	7	3

Table 2: Hand check of the questions. The built-in questions are our baseline: a person wrote and checked them.

4.2 Test day

2 classmates (users A and B; we do not write names) tested the extension on 28 October 2026. Each user:

1. installed it alone from the zip file;
2. used TikTok, Shorts or Reels as usual for 20 minutes;
3. did the quiz when the tabs closed;
4. answered "Was it fun? Rate from 1 to 5."

The users got an AI key, so on test day they saw AI questions.

Is this the real date of your test day? Did anything change in the plan? Write what really happened.

5 Results

User	tabs closed	quizzes (passed)	right answers (on average)	fun (1–5)
A	3 of 3	3 (2)	8 / 10	4
B	3 of 3	7 (5)	9 / 10	4
Total	6 of 6	10 (7)	9 / 10	4

Table 3: Test day, one row per user. Limit: 20 minutes. All users had AI questions. Pass rate: 3 of 4 quizzes.

6 Limitations

- It works only in Chrome on a laptop. Phones cannot use Chrome extensions, and most short videos are watched on phones.
- A user can turn the extension off, or open the videos in another browser.
- The AI questions can be wrong. We checked only a few.

7 Ethics

The users were volunteers. Before the test we told them what the extension does, that it takes about 20 minutes, that we write only numbers and no names, and that they can remove it at any time. The counters stay on the user's computer, and the users sent us only their result line. The AI key is personal and is not in our code or on GitHub. Figure 1 was made with an image model. We used AI coding assistants to write the code; we checked all numbers ourselves. Add anything else, for example who checked the questions and how you kept your key safe.

8 Conclusion

We built Cure Brainrot, a Chrome extension that closes short-video tabs after a daily limit and gives 20 more minutes only after a short school quiz with AI or built-in questions. In a one-day test with 2 classmates it closed 6 short-video tabs, they passed 7 of 10 quizzes with 8.5 of 10 right answers and rated it 4 of 5 for fun, and in our hand check of 30 AI questions 4 were unclear and none were wrong. Next we would make a phone version, check many more questions in all three languages, and run a longer test with more users and a group without the extension.

Acknowledgments

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References

[1] Francesco Chiossi, Luke Haliburton, Changkun Ou, Andreas Butz, and Albrecht Schmidt. Short-form videos degrade our capacity to retain intentions: Effect of context switching on prospective memory. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. ACM, 2023. doi: 10.1145/3544548.3580778.

[2] Chrome for Developers. Chrome extension API reference: tabs, storage, alarms. https://developer.chrome.com/docs/extensions/reference/api, 2026. Accessed 27 October 2026.

[3] Chrome for Developers. Extensions / Manifest V3. https://developer.chrome.com/docs/extensions/develop/migrate/what-is-mv3, 2026. Accessed 27 October 2026.

[4] Ulrik Lyngs, Kai Lukoff, Petr Slovak, Reuben Binns, Adam Slack, Michael Inzlicht, Max Van Kleek, and Nigel Shadbolt. Self-control in cyberspace: Applying dual systems theory to a review of digital self-control tools. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)*. ACM, 2019. doi: 10.1145/3290605.3300361.

[5] Christian Montag, Haibo Yang, and Jon D. Elhai. On the psychology of TikTok use: A first glimpse from empirical findings. *Frontiers in Public Health*, 9:641673, 2021. doi: 10.3389/fpubh.2021.641673.

[6] Henry L. Roediger and Jeffrey D. Karpicke. Test-enhanced learning: Taking memory tests improves long-term retention. *Psychological Science*, 17(3):249–255, 2006. doi: 10.1111/j.1467-9280.2006.01693.x.